WEEK 5 · LECTURE L2 · UNIT II

Learning Systems: From LMS to Learning Experience

How adults learn, what a learning platform can and cannot do, what makes learning change behavior, and how to tell whether training worked.

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AFTER THIS LECTURE YOU SHOULD BE ABLE TO

What this lecture prepares you for

1. Apply andragogy, Kolb’s cycle and self determination theory to digital learning design, and reject the learning styles myth with evidence.
2. Explain the forgetting curve and desirable difficulties: spacing, retrieval, interleaving, segmenting.
3. Distinguish LMS, LXP and skills platforms, using the Infosys Lex case.
4. Evaluate training with Kirkpatrick and its critics, transfer of training, 70:20:10 and training ROI.

ONE SOURCE

The same theories as the in class lecture and the course hub, without the simulation data used in class. Course outcomes: CLO 2 and CLO 4.

COMMIT BEFORE THEORY

Commit to an answer first

A bank’s new learning platform tags every employee as a visual, auditory or kinesthetic learner and serves content to match. What does the research say?

A

It raises learning substantially, because content matches each style

B

There is little evidence that matching content to a learning style improves learning

C

It works for adults but not for students

D

It works only for technical subjects

EXAM LENS

The answer is B (Pashler et al., 2008).

HOW ADULTS LEARN

Two theories set the design brief

Knowles: andragogy (1984)

Adults need to know why; they direct their own learning; experience is their main resource; they learn around real problems.

Kolb: the experiential cycle (1984)

Concrete experience → reflective observation → abstract conceptualization → active experimentation. A content library serves only one stage. Kolb also named four styles (diverger, assimilator, converger, accommodator).

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Design so every learner passes through the whole cycle, with varied formats; do not sort learners by style.

PASHLER, MCDANIEL, ROHRER AND BJORK (2008)

Learning styles: an idea without evidence

The meshing hypothesis

Teaching each person in their preferred style (visual, auditory, kinesthetic) improves learning.

What a fair test needs

People sorted by style, randomly given matched or mismatched teaching, then the same test; the result must cross over.

What they found

Virtually no studies met that design, and the few that did contradicted the hypothesis.

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Match the method to the content and the task, not to a claimed learning style. Citing this earns credit for criticality.

RECENT EVIDENCE · NEWTON AND SALVI (2020)

Across 37 studies of 15,405 educators in 18 countries, 89.1% said they believe in matching teaching to learning styles; teaching them the evidence cut that belief by 37% on average. Most studies used small convenience samples.

SELF DETERMINATION THEORY

Why points and badges can backfire

Three basic needs

Deci and Ryan (1985); Ryan and Deci (2000): autonomy, competence and relatedness sustain intrinsic motivation.

Overjustification

Deci, Koestner and Ryan (1999), meta analysis: expected tangible rewards undermined intrinsic motivation for interesting tasks; informational feedback did not.

Design consequences

Choice of path (autonomy), visible skill growth (competence), learning with peers (relatedness).

RECENT EVIDENCE · SAILER AND HOMNER (2020)

A meta analysis finds small positive effects of gamification on cognitive (g = 0.49), motivational (g = 0.36) and behavioral (g = 0.25) learning outcomes. Only the cognitive effect was stable in the most rigorous studies.

MAKING LEARNING STICK

The forgetting curve and desirable difficulties

Principle	Finding	Design implication
Forgetting	Memory fades fast without review (Ebbinghaus, 1885)	Plan reviews, not single sessions
Spacing	Spread practice beats massed practice (Cepeda et al., 2006)	Short refreshers on days 2, 7 and 21
Retrieval practice	Testing beats rereading (Roediger and Karpicke, 2006)	Quiz before and after content
Interleaving	Mixed problems beat blocked ones (Rohrer and Taylor, 2007)	Mix related tasks in practice
Segmenting	Short learner paced parts help novices (Mayer, 2009)	One task per short clip

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Bjork (1994) called these “desirable difficulties”: they slow apparent progress but improve long term retention. Short is not the active ingredient; spacing is. A 40 hour course chopped into clips taken in one sitting is still massed practice.

RECENT EVIDENCE · CARPENTER, PAN AND BUTLER (2022)

Spacing and retrieval practice are among the most reliable findings in learning research, yet learners underuse both, partly because of false beliefs about learning and because the techniques feel harder.

PLATFORMS

LMS, LXP and skills platform: three logics

	LMS	LXP	Skills platform
Organizing unit	Course	Content experience	Skill
Who chooses	HR assigns	Learner browses; platform recommends	Role gap drives
Main metric	Completion	Engagement	Proficiency gain
Best use	Compliance, certification, audit trails	Self directed professionals, discovery	Frontline reskilling
Risk	Measures activity	Popular content, not needed skills	Needs a skills taxonomy

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Most firms need both an LMS for what must be learned and an LXP for what people choose to learn. The term LXP was popularized by Josh Bersin around 2017. Mobile first, vernacular micro modules fit a frontline that cannot leave the floor.

CASE · INFOSYS (2018)

Infosys Lex: digital learning at scale

What it is

Infosys's in house learning platform, launched in April 2018: mobile first, with recommendations and peer created content.

Scale

Used by more than 1,14,000 Infosys employees within months of launch.

Then sold outside

In September 2018 Infosys began offering the platform to client firms as Infosys Wingspan.

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Usage at scale shows reach (Kirkpatrick levels 1 and 2 at best). Did learning change project behavior and results? Ask which organizing unit Lex chose and what that made measurable; then ask what would not transfer to a frontline, multilingual, smartphone only workforce.

Infosys press release, 20 September 2018.

KIRKPATRICK (1959); KIRKPATRICK AND KIRKPATRICK (2016)

Did the training work? Four levels

Level	Question	Typical evidence
4 · Results	Did organizational outcomes change?	Attrition, service quality, productivity
3 · Behavior	Do people do the job differently?	Observed use, transactions, customer handling
2 · Learning	Did knowledge and skill improve?	Assessment scores
1 · Reaction	Did learners find it useful and engaging?	Feedback forms

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Completion rates measure none of these: they record attendance. The New World Kirkpatrick model starts at level 4: decide the result first, then design backward; success at one level does not guarantee the next (the missing link).

THE CRITIQUE

The four levels are not a causal chain

Alliger et al. (1997)

Meta analysis: reactions correlated only weakly with learning and with on the job behavior. Judgments of usefulness did better than enjoyment.

Holton (1996)

The model is a taxonomy, not a theory: it names outcomes without explaining why one leads to another; context and motivation are left out.

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A high level 2 score licenses no claim about level 3. Behavior must be measured directly.

BALDWIN AND FORD (1988); BLUME ET AL. (2010)

Transfer of training: the missing causal model

Training inputs

Trainee ability and motivation; training design (practice, relevance); work environment (support, opportunity to use).

Learning and retention

What is learned and retained.

Transfer

Generalization to new tasks, and maintenance over time. Work environment acts on transfer directly: no chance to practice, no transfer.

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Blume et al.: work environment support and motivation relate to transfer. A module cannot supply the manager who lets you use what you learned: transfer is an organizational outcome, not a course outcome.

70:20:10

Popular, but weakly evidenced

The claim

About 70% of learning comes from work experience, 20% from others and 10% from formal courses.

Its origin

Center for Creative Leadership research on how successful managers learned (McCall, Lombardo and Morrison, 1988), popularized by Lombardo and Eichinger (1996). The ratio comes from executives' self reports, not measurement.

Its value

A reminder to design learning into the job, with coaching and stretch tasks; an LMS captures only the smallest part.

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Use 70:20:10 only with its evidence caveat; presenting it as a research finding loses marks. For a frontline associate, the shift and the buddy are most of the curriculum.

THEORY IN DEPTH · PHILLIPS (1997)

Add a fifth level: return on investment

The core idea

Phillips extended Kirkpatrick with a fifth level that compares monetary benefits with costs.

How it works

Isolate the training effect, convert it to money, and divide by the full cost.

Limits and critique

Isolating the effect is hard; ROI figures are often more precise than the evidence allows.

IN PRACTICE

Before claiming ROI for a learning platform, show that behavior changed; completion alone is not evidence.

EXAM LENS

Critique ROI through the counterfactual problem from Week 3: what would have happened without the training?

ADAPTIVE LEARNING AND A LEARNING PATH

From the teaching note, and an applied path

Adaptive learning

Systems adjust difficulty, pace and sequence to each learner, using the zone of proximal development, spaced repetition, interleaving and retrieval practice.

A store associate path (Course framing)

Months 1 to 3: using the self service app for one’s own pay and leave. Months 4 to 9: helping customers with digital orders and returns. Months 10 to 18: the store’s digital helper for colleagues. Measure at level 3: behavior, not completion.

National infrastructure

NASSCOM and MeitY’s FutureSkills Prime offers courses and a skills framework for emerging technology roles; it is aimed at the technology workforce, not basic digital fluency for store staff.

APPLY

A learning platform for 20,000 store staff

Teaching case (Course framing)

A national electronics retailer with 20,000 store staff buys an LXP. The vendor promises learning style personalization and 40 hour courses split into five minute clips. After six months, 82% of staff have completed the product course, but store managers report no change in how staff explain products to customers.

Work it through

1. Which Kirkpatrick level does 82% completion show?
2. Is five minute clips taken in one sitting micro learning?
3. What would you say about learning style personalization?
4. Where in Baldwin and Ford’s model is transfer failing?

COMMON CONFUSIONS

Tempting readings, and better ones

A tempting reading	A better reading
Completion rate shows training worked	Completion is activity; learning, behavior and results are levels 2 to 4
Matching content to learning styles improves learning	Pashler et al. (2008) found no good evidence for it
Short modules are micro learning	Spacing, not shortness, improves retention
Points and badges always motivate	Expected tangible rewards can undermine intrinsic motivation
70:20:10 is a research finding	It comes from self reports; treat it as a reminder

COMPARE AND EVALUATE

Lenses on digital learning

Lens	Use it to	Watch out for
Kolb (1984)	Cover the whole learning cycle	Its styles are not evidence based
Self determination theory	Design for motivation	Rewards can backfire
Desirable difficulties	Design for memory	Learners mistake ease for learning *
Kirkpatrick	Name evaluation levels	A taxonomy, not a causal chain
Baldwin and Ford	Explain transfer	Work environment is outside the course
Phillips ROI	Express value in money	Isolation and attribution

* Course framing.

PRACTICE FOR THE END TERM EXAMINATION

Questions to rehearse (1 of 2)

A Chennai IT services firm reports that 95% of employees completed its new AI skills program on its learning platform. Using Kolb, Kirkpatrick and Baldwin and Ford, evaluate whether the program succeeded and propose how you would find out. **20 marks**

A strong answer: separates completion from learning, behavior and results; checks the design covers all four Kolb stages; identifies the work environment as the main transfer risk; proposes level 3 and 4 measures and a limit such as attribution.

“Completion rates are the most misleading number in corporate learning.” Critically evaluate using Kirkpatrick, its critics and transfer of training research. **20 marks**

A strong answer: explains the four levels; uses Alliger et al. and Holton; applies Baldwin and Ford and Blume et al.; proposes behavior level measures.

PRACTICE FOR THE END TERM EXAMINATION

Questions to rehearse (2 of 2)

Using learning science, design principles for a mobile learning program for frontline employees, and evaluate the Infosys Lex model’s transferability. **20 marks**

A strong answer: applies spacing, retrieval, segmenting and self determination theory; distinguishes LMS, LXP and skills platforms; critiques transfer from graduates on laptops to frontline staff.

Why might points and badges reduce learners’ motivation? **10 marks**

A strong answer: explains self determination theory and the overjustification evidence.

CARRY THESE INTO THE EXAMINATION

Four things to remember

1. **Learning systems should change behavior, not count completions.**
   Measure transfer directly.

2. **Design for how memory works.**
   Space, retrieve, interleave and segment.

3. **Reject learning styles; keep the whole cycle.**
   Match method to task, not to a claimed style.

4. **Evidence before ROI.**
   No behavior change, no return to claim.

SET READINGS · FROM THE LESSON PLAN

Read for this lecture

- Case: Infosys Lex, digital reskilling at scale.
- NASSCOM FutureSkills (latest), selected sections.
- Scully-Russ, E., & Torraco, R. J. (2020), pp. 83 to 84.

NOTE

The other works named in this lecture are theory sources for reference, not set readings.

THEORY SOURCES · FOR REFERENCE

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